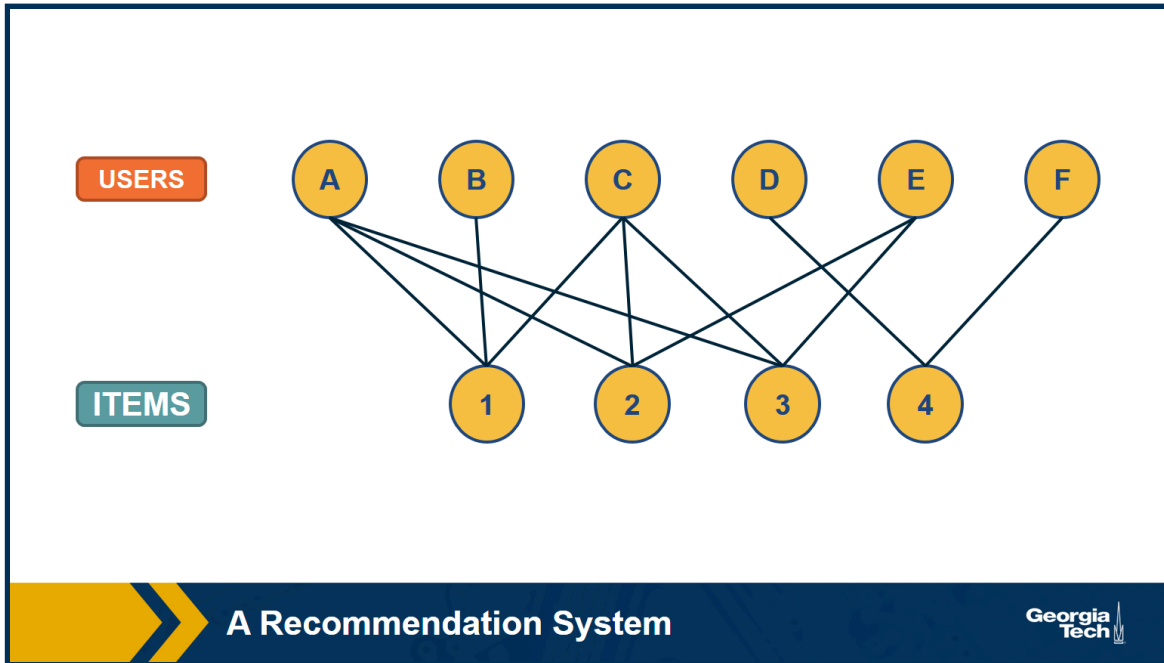


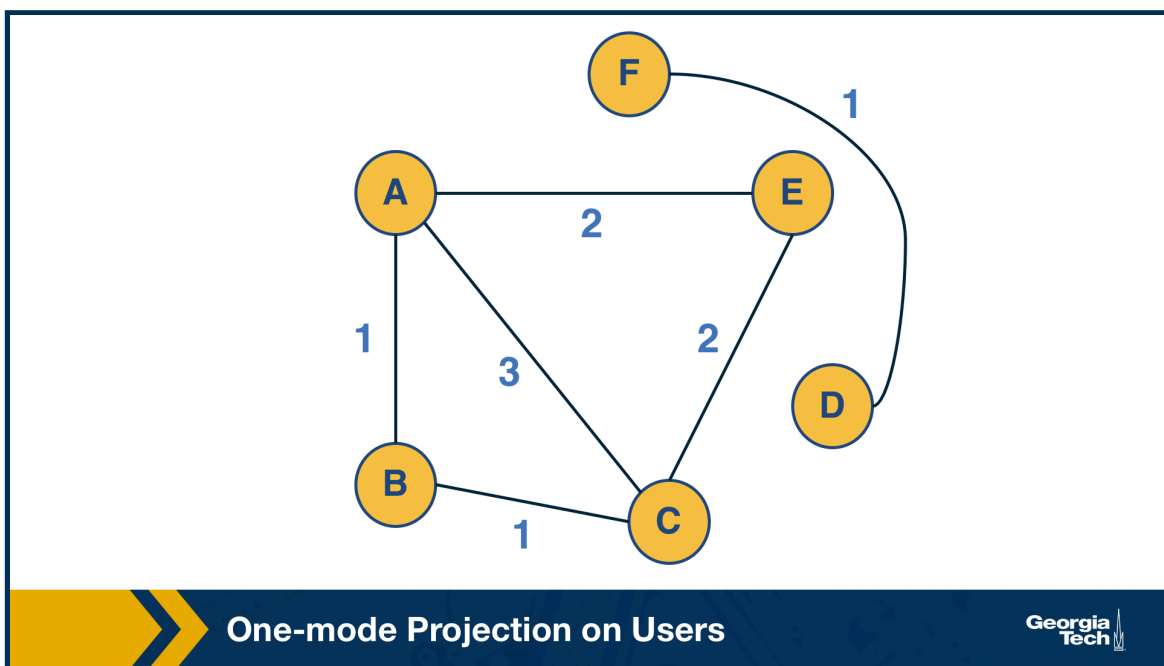
L2: A Recommendation System as a Bipartite Graph

Let's close this lesson with a practical application of bipartite graphs.

Suppose you want to create a **“recommendation system”** for an e-commerce site. You are given a dataset that includes the items that each user has purchased in the past. You can represent this dataset with a bipartite graph that has users on one side and items on the other side. Each edge (*user*, *item*) represents that that user has purchased the corresponding item.

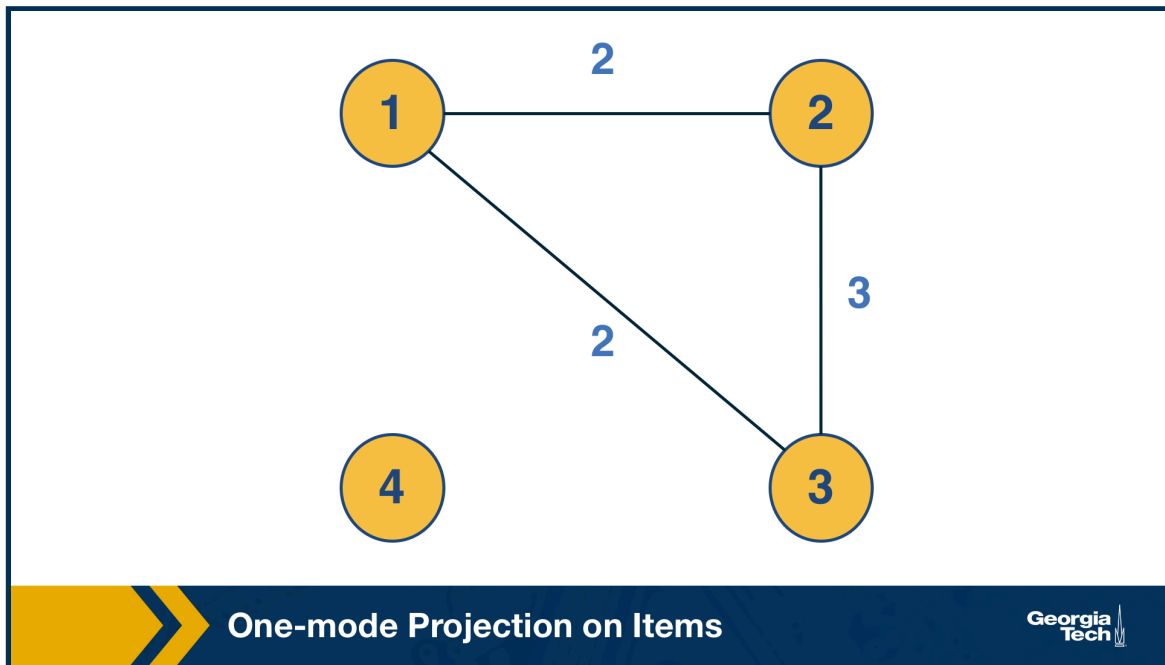


How would you find users that have similar item preferences? Having such **“similar users”** means that we can give recommendations that are more likely to lead to a new purchase.



This question can be answered by computing the **“one-mode projection”** of the bipartite graph onto the set of users. This projection is a graph that includes only the set of users – and an edge between two users if they have purchased at least one common item. The weight of the edge is the number of items they have both purchased.

How would you find items that are often purchased together by the same user? Knowing about such “*similar items*” is also useful because we can place them close to each other or suggest that the user considers them both.



This can be computed by the “*one-mode projection*” of the bipartite graph onto the set of items. As in the previous projection, two items are connected with a weighted edge that represents the number of users that have purchased both items.